Study Characteristics Table

Abbreviations:

COPD = chronic obstructive pulmonary disease
HIV/AIDS = human immunodeficiency virus/acquired immunodeficiency disease
TB = tuberculosis
ACE-I = angiotensin-converting-enzyme inhibitors
HAART = highly active anti-retroviral therapy
ART = antiretroviral therapy
MM = multimorbidity
PDC = proportion of days covered
MPR = medication possession ration
CMG = cumulative mean gap interval
MEMS = medication event monitoring system
VAS = visual analogue scale
MMAS = Morisky Medication Adherence Scale [1,2]
MARS = Medication Adherence Rating Scale[3]
CASE Index = Center for Adherence Support Evaluation [4]
HADS = Hospital Anxiety and Depression Scale[5]
BDI = Beck Depression Inventory [6]
PHQ = Patient Health Questionnaire [7]
MINI = Mini International Neuropsychiatric Interview [8]
CES-D = Centres for Epidemiologic Studies Depression Scale[9]
GDS = Geriatric Depression Scale [10]
SD = standard deviation
IQR = inter quartile range
95% CI = 95% confidence interval
RCT = randomised controlled trial

Note 1: where *n* non-adherent was not reported, *n* was calculated from the reported % non-adherent.
Note 2: where *n* female was not reported, *n* was calculated from the reported % female/male.
Note 3: pooled mean (SD) age was calculated where mean (SD) age was reported for two groups.
Note 4: participants with depression had a clinical diagnosis unless a self-report measure is noted.

Study (country)	Setting	Design	Chronic Conditions	Sample Size	Adherence Measure	Non-adherence definition	N (%) non-adherent/adherence outcome reported
Agarwal 2009 (USA)[11]	Healthcare database	Retrospective cohort	- Treated with antihypertensive - Treated with lipid-lowering agent	15400	MPR (antihypertensive (AH); lipid-lowering (LL) agent)	MPR < 0.8 for one or more medications	2 months, 6 months, 12 months: Cohort 1 (AH before LL): 40.6%, 67.3%, and 68.7% Cohort 2 (LL before AH): 55.0%, 69.2%, and 69.0% Cohort 3 (AH/LL simultaneously): 24.8%, 65.6%%, and 66.0%
Albrecht 2017 (USA)[12]	Medicare database	Retrospective cohort	- COPD - Depression	16075	PDC (COPD maintenance medications; antidepressants)	PDC < 0.8 (also report PDC = 0)	Person months: COPD: 78.8% Antidepressants: 45.2%
Albrecht 2016 (USA)[13]	Medicare database	Retrospective cohort	- COPD - Depression	31033; MM = 9593	PDC (COPD maintenance medications)	PDC < 0.8 (also report other categories)	7674 (80%)
An 2013 (USA)[14]	Administrative claims and electronic medical records	Retrospective observational	- Hypertension - Type 2 diabetes	2334	PDC (diabetes medications and hypertension medications simultaneously)	PDC < 0.8 for both drugs simultaneously	1671 (71.6%)
Applebaum 2009 (USA)[15]	HIV/AIDS primary care	Baseline data from longitudinal study	- HIV - Substance related disorder - Psychiatric disorder	67	Medication Adherence Questionnaire	Adherence < 95%	20 (29.85%)
Applebaum 2009 (USA)[16]	Healthcare clinics	Cross-sectional (data from RCT screening)	- HIV - History of injection drug use/dependence	108	MEMS (HAART); AIDS Clinical Trials Group Adherence Questionnaire (HAART)	Report mean	Mean MEMS = 68.21% Mean self-report = 93.82%
Axon 2016 (USA)[17]	Veterans health administrative databases	Retrospective cohort	- Type 2 diabetes - Depression	740197; MM = 88895	MPR (diabetes medications)	MPR < 0.8	28143 (31.7%)
Baca-Garcia 2009 (USA)[18]	University hospital	Cross-sectional	- Current bipolar major depressive episode - Alcohol use disorder	97; MM = 39; analysed n = 36	Self-report (antidepressants)	Report mean (SD) median global compliance	Mean (SD) = 90.7 (14.0)
Badiee 2012 (USA)[19]	Ongoing studies at university research centre	Cross-sectional	- HIV - Bipolar disorder I or II	74; MM = 43	MEMS (protease inhibitor or related) AIDS Clinical Trials Group (ACTG) 4-day Adherence to	MEMS < 90% ACTG < 95%	MEMS: 22 (51%) ACTG: 6 (13%)

					Antiretrovirals Questionnaire (antiretroviral medications) Visual Analogue Scale (protease inhibitor)	Visual Analogue Scale < 95%	Visual Analogue Scale: 22 (52%)
Baik 2012 (USA)[20]	Medicare database	Retrospective cohort	- Depression - Heart Failure	552956; MM = 22118; analysed n = 13981	MPR (heart failure medications; antidepressants)	Report mean	No coverage group: Antidepressants: Mean MPR = 0.56 Heart failure medications: Mean MPR = 0.89 Generic coverage group: Antidepressants: Mean MPR = 0.60 Heart failure medications: Mean MPR = 0.90
Banegas 2018 (USA)[21]	Healthcare database	Retrospective cohort	- Treated with statin - Newly diagnosed cancer (breast / prostate / colorectal)	10177	PDC (statins)	Partial adherence = PDC 0.20-0.79 Non-adherence = PDC < 0.2	Year 1: PDC < 0.20 = 837 (8.2%) PDC 0.20-0.79 = 2280 (22.4%) Year 2: PDC < 0.20 = 887 (8.7%) PDC 0.20-0.79 = 2155 (21.2%)
Basak 2014 (USA)[22]	Healthcare database	Longitudinal observational	- Type 2 diabetes - Hypertension and/or dyslipidaemia or angina	6043	PDC	Report mean (SD)	Report according to medication combinations per quarter. Range = 24.43 (39.10) for antianginal medications to 86.40 (23.27) for hypertension medications.
Batchelder 2014 (USA)[23]	Community and hospital	Cross-sectional	- HIV - Type 2 diabetes	62	VAS (antiretrovirals; diabetes medications; insulin injections where appropriate)	VAS < 100%	ARV: 32 (51.6%) Antidiabetes: 39 (63%) Of 32 participants reporting <100 % to HIV medication, 97 % (31) reported < 100 % to diabetes medication
Bauer 2013 (USA)[24]	Healthcare database	Observational new prescription cohort follow-up	- Type 2 diabetes - Depression	1366	Pharmacy utilisation data (antidepressants)	Primary non-adherence: no dispensing of index prescription within 60 days of index prescription date. Non-persistence: dispensing of index	Primary non-adherence: 52 (3.8%) Early non-persistent: 432/1314 (32.9%) 2nd stage non-persistent: 83/1314 (6.3%)

						prescription once, no additional dispensing within period defined by number of days' supply of medication dispensed plus a grace period equal to 100% of prior days' supply.	
Beadles 2015 (USA)[25]	Medicaid database	Retrospective cohort	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	depression = 9303; hypertension = 12595; diabetes = 6409; hyperlipidaemia = 9263	PDC (target medication)	Report mean (SD)	Mean (SD): Depression = 40% (43%) Hypertension = 50% (43%) Diabetes = 49% (43%) Hyperlipidaemia = 47% (43%)
Benner 2009 (USA)[26]	Healthcare database	Retrospective claims analysis	- Treated with antihypertensives - Treated with lipid-lowering therapy	5759	PDC (for combination therapy, reported according to prescription burden)	PDC < 80%	By number of other medications: 0 = 59.0% 1 = 65.5% 2 = 69.9% 8 = 79.9% 9 = 74.5% 10/+ = 79.9%
Berg 2018 (Denmark)[27]	Hospital	Cross-sectional	- Cardiac Disease - Depression/anxiety (HADS)	14239; MM (anxiety) = 1978; MM (depression) = 982	Single item self-report (cardiac medication)	Forgetting to take prescribed medication more than once a week among patients who took medication prior to hospital admission	Anxiety group = 113 (7%) Depression group = 70 (8%)
Bifftu 2015 (Ethiopia)[28]	Hospital outpatient	Cross-sectional	- Epilepsy - Depression (BDI-II)	405; MM = 183	Single item self-report (antiepileptic medications)	Not defined (reported as number with difficulties of adherence to antiepileptic drugs)	81 (44.3%)

Blashill 2015 (USA)[29]	HIV primary care clinics	Baseline data of RCT screening	- HIV - Depressive symptoms (PHQ-2)	333	MEMS (antiretrovirals)	MEMS < 80%	191 (57.4%)
Bogner 2014 (USA)[30]	Primary care	Longitudinal	- Type 2 diabetes - Depression	180	MEMS (hypoglycaemic agents and antidepressants) Brief Medication Questionnaire (BMQ) (hypoglycaemic agents and antidepressants)	MEMS < 80% BMQ < 80%	Oral hypoglycaemic agent: MEMS = 103 (57.22%) Self-report = 34 (18.89%) Antidepressants: MEMS = 111 (61.67%) Self-report = 47 (26.11%)
Cabrera-Marroquin 2014 (Mexico)[31]	Tertiary care (ambulatory)	Observational cross-sectional	- Rheumatoid arthritis; - Major depressive disorder	78; MM = 24	Concordance with Therapy self-report measure of adherence and persistence (RA medication)	Based on combinations of item scores	10 (41.7%)
Calderón-Larrañaga 2016 (Spain)[32]	Primary care	Cross-sectional	-Hypertension -Physical and/or mental comorbidity	113,397; MM with physical co-morbidity = 103913; MM with mental comorbidity = 20349	MPR (antihypertensives)	MPR < 80%	+ physical comorbidity = 20868 (20.08%) + mental comorbidity = 4387 (21.56%)
Calip 2013 (USA)[33]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with statins	1393	Adherence: MPR (statins) Persistence: discontinuation rate (statins)	MPR < 0.8 Discontinuation rate (DR): gap ≥ 90 days between end of previous days' supply and subsequent dispensing of statin medication	Cancer treatment period: MPR < 0.8 = 666/1384 (48.1%) Mean (SD) DR = 0.5 (0.6) Year +1: MPR < 0.8 = 572/1129 (50.7%) Mean (SD) DR = 1.2 (1.3) Year +2: MPR < 0.8 = 672/1040 (64.6%) Mean (SD) DR = 1.2 (1.2) Year +3: MPR < 0.8 = 477/959 (49.7%) Mean (SD) DR = 1.2 (1.2)
Calip 2015 (USA)[34]	Healthcare database	Retrospective cohort	- Diabetes mellitus - Breast cancer	509; analysed n = 499	MPR (oral diabetes medications)	MPR < 0.8	376 (75.4%)

Calip 2017 (USA)[35]	Healthcare database	Retrospective cohort	- Breast cancer survivors - Treated with at least one of antihypertensives, oral antidiabetes medications or statins	2308	MPR (antihypertensive or statins or oral antidiabetic medications)	MPR < 0.8	Antihypertensives = 666/1779 (37.4%) Oral diabetes medications = 379/499 (76.0%) Statins = 416/1072 (38.8%)
Cantudo-Cuenca 2014 (Spain)[36]	Hospital pharmacy service	Prospective observational	- HIV - Comorbidity(s)	594; MM = 431	MMAS-4 Pharmacy dispensing records: Number of medications dispensed/number of medications prescribed x 100	MMAS-4 < 4 Pharmacy data < 90%	129 (29.9%) (results show pharmacy data; authors reported no difference in adherence between both measures)
Casaletto 2016 (USA)[37]	Not reported	Cross-sectional	- HIV - Bipolar I or II	50	MEMS (bipolar medications)	MEMS < 90%	28 (56.0%)
Caughey 2013 (Australia)[38]	Veterans' Affairs database	Retrospective cohort	- Diabetes - Depression	29710; MM = 7171	Pharmacy dispensing data – treatment discontinuation (antidiabetic medications)	Gap between two dispensings of more than three times the duration	2868 (40%) Median (95% CI) = 1.81 (1.72-1.94)
Cha 2012 (USA)[39]	Not reported	Cross-sectional using baseline data from intervention	- Heart failure - Diabetes mellitus	116; MM = 46	MMAS-8 (heart failure medications) MEMS (ACE inhibitor and diuretic)	Report mean (SD): MMAS-8 range = 1-12; higher score = more adherent	MMAS-8 mean (SD) = 10.6 (1.6) MEMS: correct doses: ACE-I = 88.0 (16.6) Diuretics = 89.0 (17.0) MEMS: correct days: ACE-I = 81.7 (22.8) Diuretics = 81.1 (22.6)
Chou 2017 (USA)[40]	Healthcare database	Retrospective cohort	- Depression - Breast cancer (newly diagnosed)	2284; MM = 1142	PDC (antidepressants)	PDC < 80%	494 (43.3%)
Cicolini 2016 (Italy)[41]	Primary care	Cross-sectional	≥ 2 chronic condition from cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	567	MMAS-8 (Italian Version)	MMAS-8 score between 1-5.75 (defines "low adherence")	263 (46.4%)

Crowley 2013 (USA)[42]	Community practice	Cross-sectional (baseline data from RCT)	- Type 2 diabetes - Hypertension	377	MMAS-4 (hypertension medications; diabetes medications)	Score ≤ 2 on any item	Hypertension = 110 (29.2%) Diabetes = 162 (43.0%)
Crowley 2015 (USA)[43]	Tertiary care health system	Cross-sectional (baseline data from RCT)	- Acute myocardial infarction in previous 3 years - Hypertension	405	MMAS-4 (general adherence)	Response of "strongly agree," "agree," "don't know," or "refused" for any statements	173 (42.7%)
Cruess 2012 (USA)[44]	Physician outpatient clinics	Prospective cohort	- HIV - Treated with psychotropic medication	324; MM = 106	Monthly unannounced pill counts (HIV medication; psychiatric medication)	Report mean (SD)	HIV = 84.09% (19.26%) Psychiatric = 79.86% (20.88%)
Cziraky 2015 (USA)[45]	Healthcare database	Retrospective cohort	- Acute coronary syndrome or unstable angina - Type 2 diabetes	Year 1 = 85097; MM = 22588 Year 3 = 35540; MM = 9160	MPR (cardiovascular medications) Treatment discontinuation (cardiovascular medications)	MPR: not reported Treatment discontinuation: gap in treatment of 45 days or longer in each medication based on days of supply	Oral diabetes medication: Year 1 = 49.3% Year 3 = 46.8% Insulin Year 1 = 77.4% Year 3 = 78.9% Additional medications reported in original paper.
Daniali 2018 (Iran)[46]	Government healthcare centres	Cross-sectional	- Chronic condition - Depression (PHQ-9)	483; MM = 79	4-item self-report	Not reported	40 (50.6%)
Daniels 2016 (USA)[47]	Primary care	Pilot intervention	≥ 2 chronic conditions	20	MMAS-8	Report mean (SD) MMAS-8 score > 2 (defined as "low adherence"); lower score = lower non-adherence	Mean (SD) = 4.65 (2.37)
Davies 2015 (USA)[48]	Tertiary care Veterans Affairs Medical Centre (patients receiving primary care)	Retrospective chart review	- HIV/AIDS - Diabetes mellitus	56	MPR (ART)	MPR < 95%	34 (60.7%)
Davis 2011 (USA)[49]	Clinical centres	Data from RCT	- Hypertension	1065	Not reported (antihypertensives)	Report mean (SD)	Lower BP goal group: Achieved goal: 92.6% (7.7%)

			- Chronic kidney disease				Above goal: 83.0% (14.7%) Usual BP goal group: At goal: 91.4% (8.8%) Below goal: 90.8% (13.2%) Above goal: 78.5% (19.1%)
DeLeon 2015 (USA)[50]	Primary care	Retrospective cohort	- Diabetes - Hypertension	20085; MM = 3827	MPR (diabetes medications, hypertension medications, lipid controlling medications)	MPR < 80%	Diabetes: 1056/1679 (62.9%) Hypertension: 1084/1923 (56.4%) Lipid controlling: 928/1295 (71.7%)
Desai 2014 (USA)[51]	Medicaid database	Retrospective analysis	- Treatment by antipsychotic - Treatment by antidiabetic medication	3642; MM = 1821	PDC (oral antidiabetics; oral antipsychotics) Persistence (oral antidiabetics)	PDC < 0.8 Persistence: number of days of continuous access to oral antidiabetics, allowing for a 30-day gap period.	Antidiabetics: 1143 (62.8%) Mean (SD) persistence = 180.8 (124.8) Antipsychotics: 689 (37.8%)
Dhamane 2017 (USA)[52]	Healthcare database	Retrospective cohort	- COPD - Treatment by ≥ 1 of antianxiety, anticoagulants, antidepressants, antihyperglycaemics, antihypertensives, beta-blockers, bisphosphonates, calcium channel blockers, diuretics, insulin, NSAIDs, statins	14117	PDC (COPD medications; non-COPD medications)	PDC < 0.8 Report mean PDC for non-COPD medications	COPD medications: 11177 (79.2%) non-COPD medications mean: antianxiety = 0.39 anticoagulants = 0.55 antidepressants = 0.65 antihyperglycaemics = 0.70 antihypertensives = 0.71 beta-blockers = 0.70 bisphosphonates = 0.54 calcium channel blockers = 0.68 diuretics = 0.65 insulin = 0.53 NSAIDs = 0.33 statins = 0.67
Domino 2014 (USA)[53]	Administrative claims databases	Retrospective secondary data analysis	≥ 2 chronic conditions from asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	188531	PDC	Report mean PDC	Diabetes = 0.29 Hyperlipidaemia = 0.29 Hypertension = 0.43 Depression = 0.33 Schizophrenia = 0.46

Druss 2010 (USA)[54]	Community mental health centre	Pilot RCT	- ≥ 1 chronic medical condition - Severe mental illness	80	MMAS-4 (chronic medical condition)	Report mean (SD) score from 0 (no problems with adherence) to 4 (more problems with adherence)	Mean (SD) = 1.5 (1.3)
DuChane 2014 (USA)[55]	Pharmacy claims database	Retrospective cohort	- HIV - Comorbidity (treated with angiotensin-converting enzyme (ACE) or angiotensin receptor blocker (ARB) or statin)	ACE/ARB group = 2968; MM = 1484; Statins group = 2744; MM = 1372	PDC	ART medications: PDC < 95% Other medications: PDC < 80%	HIV + ACE/ARBs = 1118 (37.7%) HIV + statins = 936 (34.1%) Mean (SD) ART = 83.9 (24.3)
Eton 2017 (USA)[56]	Primary care	Cross-sectional	≥ 2 chronic conditions requiring self-management	332 (analysed n = 311)	Single-item self-report	Response "usually /sometimes" vs "always" re adherence to physician-recommended medications	Usually/sometimes = 51 (16.4%)
Farley 2012 (USA)[57]	Medicaid database	Retrospective cohort	- Schizophrenia - Pre-existing cardiometabolic condition(s): hypertension, hyperlipidaemia and/or diabetes	1006	PDC (antipsychotic medications; cardiometabolic medications)	PDC < 80% (unweighted mean of 3 cardiometabolic classes) Report mean (SD)	Antipsychotic (acute phase) = 89 (18.8%) Diabetes medications: acute phase = 0.90; continuation phase = 0.63; maintenance phase = 0.59 Antihypertensives: acute phase = 0.86; continuation phase = 0.71; maintenance phase = 0.66 Statins: acute phase = 0.79; continuation phase = 0.65; maintenance phase = 0.57
Feldman 2016 (USA)[58]	Not reported (recruited from outpatients, emergency room, referrals and flyers)	RCT	- Panic disorder - Asthma	48	MARS (inhaled corticosteroids)	Report mean (SD) Range = 1-5; higher score = lower non-adherence. Score 4.5+ = good adherence.	Mean (SD) = 3.8 (0.4)

Ferguson 2017 (Australia)[59]	Cardiology ward at academic medical centre	Prospective cohort	- Chronic heart failure; - Atrial fibrillation	137; analysed n = 83	MMAS-4	≥ 1 responses to non-adherence items	36 (43.4%)
Foguet-Boreu 2017 (Spain)[60]	Primary care	Cross-sectional	- Hypertension - Diabetes type 1 or type 2	27637	MPR (hypoglycaemic medications; antihypertensive medications; both combined)	MPR < 80% Report MPR ≥ 80%	Diabetes = 10898 (39.8%) Hypertension = 4322 (23.8%) Combined diseases (MPR < 80% for at least one condition) = 13593 (49.2%)
Gathright 2017 (USA)[61]	Hospital	Secondary analysis of data from longitudinal observational study	- Heart Failure - Depressive Symptoms (PHQ-9)	308; MM = 107	MEMS (up to 4 common heart failure medications)	Report mean (SD) percent of days a participant did not follow their prescribed regimen	Mean (SD) = 65.3% (27.5%)
Gaudiano 2017 (USA)[62]	Psychiatric hospital	Cross-sectional (study of baseline data from clinical trial)	- Bipolar I or II or NOS - Drug and/or alcohol use disorder	39	Medication Compliance Questionnaire (mean rating across all psychotropic medications)	Report mean (SD) Range = 0-4, higher score = higher non-adherence.	Mean (SD) = 1.2 (1.2)
Gaynes 2015 (USA)[63]	Infectious disease clinics	Data from RCT	- HIV - Depression (PHQ-9 and major depressive disorder (MINI)	304	Visual Analogue Scale (HIV/AIDS medications)	VAS score ≤ 80%	78 (25.7%) No psychiatric comorbidity = 13/54 (20.1%) ≥ 1 psychiatric comorbidity = 65/250 (26.0%)
Gentil 2015 (Canada)[64]	Community	Data from longitudinal study	- Type 2 diabetes - Depression/anxiety	301; MM = 47	MPR (oral antidiabetic medication)	MPR < 80%	18 (38.3%)
Gibson (USA) 2018[65]	Primary care	Cross-sectional	- Type 2 diabetes - Hypertension	60	Self-report during semi-structured interview	Self-report missing 1 or more days in previous week	Hypertension: 14 (23.3%) Type 2 diabetes: 16 (26.7%)
Gnanasan 2011 (Malaysia)[66]	Hospital outpatients	Feasibility study (action research)	- Diabetes mellitus - TB	35	Medication-related problems identified by clinical pharmacist (TB medications; non-TB medications)	Not reported	TB treatment: 3 (8.6%) Non-TB treatment: 10 (28.6%)
Gonzalez 2011 (USA)[67]	Methadone maintenance treatment	Cross-sectional	- HIV - History of injection drug use	91	MEMS (HAART)	MEMS < 100% doses as prescribed	73 (80.2%)

Grove 2017 (USA)[68]	Medicaid database	Retrospective cohort	- Schizophrenia - ≥ 1 chronic condition from hypertension, hyperlipidaemia, seizure disorder, chronic obstructive pulmonary disease, diabetes, asthma, or major depression	12662	PDC (antipsychotic medications)	Report mean (SD)	Mean (SD) = 45.4% (37.8%)
Gundling 2013 (Germany)[69]	Hospital	Retrospective analysis	- Diabetes mellitus - Cirrhosis	285; MM = 87; analysed n = 69	Information about compliance based on doctors', patients' and relatives' reports (report general adherence)	"e.g. concerning regular intake of medication and following the doctor's advice"	36 (52.2%)
Gupte-Singh 2015 (USA)[70]	Medicaid database	Retrospective analysis	- Asthma - Depression	3626; MM = 226	PDC (asthma controller; antidepressant medications)	Asthma controller: PDC < 50% Antidepressant: PDC < 80%	Asthma controller medications: 140 (62.0%) Antidepressants: 168 (74.3%)
Halava 2014 (Finland)[71]	Data from public sector employees	Prospective cohort	- Prescribed statins - Cardiovascular comorbidities	9285; MM = 2827	PDC (statins)	PDC < 80%	1155 (40.9%)
Han 2018 (USA)[72]	Medicare database	Retrospective cohort	- Hypertension - Chronic kidney disease	70201	PDC (antihypertensive medications)	PDC < 80%	27468 (39.1%)
Hansen 2009 (USA)[73]	Ambulatory care practice	Secondary analysis of RCT data	- Heart failure - Depression (GDS)	314; MM = 117	MEMS (heart failure medication) MMAS-4 + one-item screening questionnaire (heart failure medication)	Report mean (95% CI) MMAS-4: Range = 0-1; higher score = better adherence	MMAS-4 = 79% (71-87%) MEMS not reported
Hansen 2018 (USA)[74]	Medicaid database	Retrospective cohort	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	4223	PDC (antihypertensive medications; statins; oral antidiabetics; antipsychotic medications)	PDC < 80%	Range of non-adherence reported: Antipsychotic = 24.4% - 44.5% Antidiabetic = 39.4% - 61.5% Antihypertensive = 32.5% - 52.6% Statins = 44.5 – 63.5% Full results reported according to number of primary care prescribers

							and psychiatric specialist prescribers in original paper.
Hansen 2012 (USA)[75]	Medicaid database	Cross-sectional time-series	- Schizophrenia - Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes)	33230	PDC (oral second-generation antipsychotics; oral antidiabetics; statins; antihypertensive medications)	PDC < 0.8	Antipsychotic: 11565/33230 (34.8%) Antihypertensives: 10069/24843 (40.5%) Antidiabetics: 4869/12349 (39.4%) Antihyperlipidemics: 4192/10909 (38.4%)
Hansen 2014 (USA)[76]	Veterans' Administration Medical Centre	Retrospective cohort	- Cardiometabolic condition (hypertension, hyperlipidaemia and/or diabetes) - ≥ 1 additional chronic condition(s)	7933; Diabetes MM = 1417; Hypertension MM = 3318; Dyslipidaemia MM = 3009	CMG (diabetes medications; hypertension medications; dyslipidaemia medications)	CMG ≥ 20%	Diabetes (+ ≥ 1 other): 392 (27.7%) Hypertension (+ ≥ 1 other): 766 (23.1%) Dyslipidaemia (+ ≥ 1 other): 948 (31.5%)
Hapangama 2013 (Sri Lanka)[77]	Tertiary care psychiatric unit (inpatient and outpatient)	Cross-sectional	- Mental disorder - Substance use disorder	325; MM = 141	Interview schedule administered by senior registrar in psychiatry	Ceased medication for ≥ 2 days for any reason, without medical advice, during treatment for mental illness	91 (64.5%)
Hawke 2013[78]	Academic medical centres primarily	RCT (secondary data analysis)	- Bipolar I or II - Comorbid anxiety disorder	204; MM = 85	Single-item self-report	Report mean (SD) (possible range not reported)	Mean (SD) = 1.36 (0.53)
Hussein 2010 (USA)[79]	Healthcare database	Retrospective cohort	- Hypertension - Using a two-pill amlodipine + atorvastatin; using a two-pill CCB + statin; or using a single pill amlodipine/ atorvastatin	35430	PDC (statins; calcium channel blocker)	PDC < 80% (both medications together)	Range of non-adherence reported: 39.1 - 73.4% Full results according to status of experienced/naïve with respective medications reported in original paper.
Ihle 2017 (Switzerland)[80]	Primarily hospital	Cross-sectional	≥ 2 chronic conditions	84	Self-report items adapted from Moriskey, Green, Levine measure (specifically, assesses	Report mean (SD) from 1-5 (higher score = higher non-adherence)	Omission errors: mean (SD) = 1.39 (0.88)

					omission errors and commission errors)		Commission errors: mean (SD) = 1.12 (0.57)
Inauen 2017 (Switzerland) [81]	University hospital (outpatient) and other channels, e.g. online, pharmacy	3-wave panel study	≥ 2 chronic conditions and ≥ 2 prescribed medications	84	Non-Adherence Report, Medication Based (NAR-M) Non-Adherence Report, Diary (NARD) MARS MEMS	Report mean (SD) (lower mean = higher non-adherence)	NAR-M: T1 = 92.8 (11.1); T2 = 92.1 (12.4); T3 = 94.5 (10.5) NARM-D (no MEMS group) = 96.9 (7.9) MEMS group = 98.5 (4.1) MARS T1 = 89.4 (18.3) T2 = 90.1 (18.1) T3 = 94.5 (13.3) MEMS T1-T2 = 95.7 (3.8) T2 – T3 = 93.7 (10.4)
Iyengar 2016 (USA)[82]	Medicare database	Retrospective claims analysis	- Diabetes - Hypertension - High blood cholesterol	40632	PDC (diabetes medications, cholesterol medications, hypertension medications)	PDC < 80%	Antidiabetics = 13898 (34.2%) Antihypertensives = 13462 (33.1%) Antihyperlipidemics = 10584 (26.1%)
James 2018 (USA)[83]	Hospital	Longitudinal	- Type 2 diabetes - Early stage cancer (recently diagnosed)	56	MMAS-8 (diabetes medications)	MMAS: positive response to ≥ 2 questions	Baseline: 25 (44.6%); 3-months: 20 (35%)
Jaworski 2011 (France)[84]	University hospital psychiatric department (outpatient)	Cross-sectional	- Schizophrenia or bipolar disorder - Substance use disorder	80; MM = 40: schizophrenia + SUD= 20; bipolar + SUD = 20	Self-report number of times forgot to take medication in preceding month (psychotropic medications)	Score < 75%	Bipolar + SUD = 15 (75%); Schizophrenia + SUD = 14 (70%)
Jing 2011 (USA)[85]	Medicare and healthcare databases	Retrospective analysis	- Treatment with antihypertensive - Treatment with antidiabetic	30132	MPR	MPR < 77% for ≥ 1 antihypertensives and/or antidiabetic medications for each standalone calendar quarter	10722 (35.6%)
Joseph 2017 (India [86]	Medical college hospital & dialysis centre	Prospective observational	- Hypertension - Chronic kidney disease	120	MMAS-8 (antihypertensives)	Not reported	75 (62.5%)

	(inpatient and outpatient)						
Kacanek 2010 (USA)[87]	Physician network, newspapers, radio advertisement	Secondary analysis of data from longitudinal cohort	- HIV - Depression symptoms (CES-D and Diagnostic Interview Schedule)	225; MM= 51	For each medication, participants reported the number of days over the last seven days they had missed a dose of that antiretroviral medication.	Missing at least 5% of HAART doses in the seven days prior to study	23 (45.1%)
Kalichman 2015 (USA)[88]	AIDS community services and infectious disease clinics	Prospective	- HIV - Mental health condition	123	Unannounced pill-counts (antiretroviral medications; psychiatric medications)	Antiretrovirals: < 85% Psychiatric medications: report mean (SD)	ART: 57 (46.3%) ART mean (SD) = 77.2 (25.3) Psychiatric mean (SD) = 64.6 (36.4)
Kamal 2018 (Switzerland)[89]	University hospital infectious disease service	Prospective observational cross-sectional	- HIV - Comorbidity (≥ 1 concomitant oral medication from any therapeutic class for at least 6 months)	105	Swiss HIV Cohort Study Adherence Questionnaire (SHCS-AQ) (ART; co-treatments)	Missing ≥ 1 dose of treatment in previous 4 weeks	ART = 18 (17.1%) Co-treatments = 30 (28.6%)
Katerndahl 2012 (USA)[90]	Primary care	Cross-sectional	- Diabetes - Mild or moderate depression (PHQ-9)	106; MM = 37	Medication subscale from Diabetes Self-Care Activities Measure	Report mean (possible range = 0 – 7; lower score = lower compliance	Mean = 4.89
Katon 2009 (USA)[91]	Primary care	Prospective	- Diabetes type 1 or type 2 - Probably major depression (PHQ-9)	4117; MM = 495	CMG (hyperlipidaemia medications; hypertension medications; hyperglycaemia medications)	CMG ≥ 20%	Only reported in patients with poor glycaemic, blood pressure and lipid control: Poor diabetes control = 47/127 (37.0%) Poor hypertension control = 67/232 (28.9%) Poor lipid control = 17/44 (38.6%)
Kebede 2012 (Ethiopia)[92]	Hospital	Cross-sectional	- HIV/AIDS - TB	24	Self-report tool	Missing a dose for ≥ 1 day per month (ART; TB)	ART: 16 (66.7%) TB: 11 (45.8%)
Kenning 2015 (UK)[93]	General practice	Prospective	≥ 2 from chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and	410	Modified Moriskey Scale (6-items)	Not reported	Not reported

			depression (self-reported and/or HADS).				
Kim 2010 (South Korea)[94]	Pharmacy claims database	Retrospective cohort	- Hypertension - Psychiatric disorder	2454844; MM = 158982	Cumulative Medication Adherence (CMA) (antihypertensives)	CMA < 80%	67297 (42.3%)
Kong 2012 (USA)[95]	Medicaid database	Retrospective cohort	- HIV - Depression	7034; MM = 3286; analysed n = 3202	MPR (antiretrovirals)	Adherence < 90%	2386 (74.5%)
Kostev 2018 (Germany)[96]	General practice	Retrospective cohort	- Type 2 diabetes - Depression	12898; MM = 6449	Persistence (oral antihyperglycaemic medications)	At least 3 months without drug	2096 (32.5%)
Kreyenbuhl 2010 (USA)[97]	Veterans' Affairs administrative database	Retrospective cohort	- Type 2 diabetes - Schizophrenia or schizoaffective disorder	22014; MM = 11454	MPR (oral hyperglycaemic medications; antipsychotic medications)	MPR < 0.8	Oral antihyperglycaemics = 4925 (43%) Antipsychotic medications = 4009 (35%)
Kreyenbuhl 2011 (USA)[98]	Mental health clinics (outpatient)	Cross-sectional	- Type 2 diabetes - Serious mental illness	74; MM = 44	Brief Medication Questionnaire (oral hypoglycaemic medications)	Not taking any hypoglycaemic medications as prescribed over the past week	12 (27%)
Krivoy 2015 (Isreal)[99]	Healthcare database	Retrospective analysis	- Ischaemic heart disease - Depression	63437	MPR with addition of physician prescription data from electronic health record (antidepressants)	MPR < 20% = non-adherent MPR 20% to <50% = poorly adherent MPR 50% - <80% = moderate adherence	Non-adherence = 18212 (28.7%) Poor adherence = 10094 (15.9%) Moderate adherence = 11041 (17.4%) Total < 80% = 39347 (62.03%)
Krivoy 2017 (Israel)[100]	Healthcare database	Retrospective analysis	- Depression - Cerebrovascular accident	32361	MPR plus physician prescription data (antidepressants)	MPR < 20 % = non-adherent MPR 20-50% = poorly adherent MPR 50-80% = moderately adherent	< 20% = 8619 (26.6%) 20-50% = 5108 (15.8%) 50-80% = 5656 (17.5%) Total < 80% = 19383 (59.9%)
Krousel-Wood 2010 (USA)[101]	Managed care organisation	Prospective cohort	- Hypertension - Depression (CEDS)	2180; MM = 284	MMAS-8 (antihypertensives); MPR (antihypertensives)	MMAS-8 score < 6 ("low adherence"); MPR < 0.8	MMAS: 67 (23.6%); MPR = 104 (36.5%)

Krumme 2018 (USA)[102]	Medicare database	Retrospective cohort	≥ 2 chronic conditions, at least one of which is hypertension, hyperlipidaemia or diabetes	22805; MM = 6519	PDC (cardiovascular medication)	PDC < .80	PDC ≤ 70 = 1303 (20.0%) PDC ≤ 85 = 2865 (44.0%)
Kumar 2009 (USA)[103]	Data from longitudinal study	Cross-sectional	- HIV - Depression	1192; MM = 210	4-item self-report for past week (antiretroviral medication)	Report categories of 100%, 85%, 57-85%, <57.	Adherence < 100% = 88 (41.91%) Adherence < 85% = 67 (31.91%)
Langness 2014 (USA)[104]	Pharmacy records	Retrospective review	- HIV - Treatment with mental health medication and/or blood pressure medication	Blood pressure group = 269; Mental health group = 295	PDC (blood pressure medication; mental health medication)	PDC < 80%	Blood pressure: 92 (34.2%) Mental health: 108 (36.6%)
Lee 2011 (USA)[105]	Medicaid database	Retrospective cohort	- Serious mental illness (schizophrenia or major affective disorder) - HIV	23413; MM = 788	Continuity of medication use measures (ART and psychotropic medication)	Monthly use of any antiretroviral or psychotropic medications (divide total months patients had prescription filled by follow-up months after indexed prescription)	Percent of continuous time on medication: Psychotropic medication: 72.5% ART: 71.6%
Li 2016 (Hong Kong)[106]	Primary care	Cross-sectional	- Hypertension - Co-occurring conditions	2445; MM = 1159	MMAS-8 (Chinese version) (antihypertensives)	MMAS-8 ≤ 6	549 (47.4%) + 1 comorbidity = 339 (47.7%) + ≥ 2 comorbidities = 210 (46.9%)
Lin 2012 (USA)[107]	Primary care	RCT	- Depression (PHQ-9) - Diabetes or heart disease	214	CMG (diabetes/heart disease)	Report mean (SD)	Mean (SD): Oral hypoglycaemic = 0.83 (0.19) Antihypertensive = 0.85 (0.18) Lipid lowering = 0.83 (0.20) Antidepressant = 0.80 (0.21)
LoRe 2013 (USA)[108]	Veterans Affairs (VA) database	Retrospective cohort	- HIV - Hepatitis C virus	333	Sum of days' supply of medication dispensed over 12-week interval divided by number of days between first final fills of that interval (antiviral medications)	Pharmacy data < 70% (also report additional intervals above 70%)	Interferon: Interval 1 = 28/267 (10.5%) Interval 2 = 38/189 (20.1%) Interval 3 = 22-131 (16.8%) Interval 4 = 28-94 (29.8%) Ribavirin: Interval 1 = 46/267 (17.2%)

							Interval 2 = 55/185 (29.7%) Interval 3 = 46/127 (36.2% Interval 4 = 49/91 (53.8%)
Long 2014 (USA)[109]	Veterans Affairs medical facilities (primary care)	Cross-sectional	- Type 2 diabetes - Serious mental illness	363	MPR (diabetes medications including insulin) MMAS-4 (diabetes specific)	MPR: report mean (SD) MMAS-4: report as perfect vs non-perfect adherence	MPR mean (SD) = 0.8 (0.3) MMAS-4 = 178 (49%)
Lunghi 2017 (Canada)[110]	Healthcare database	Retrospective cohort	- Type 2 diabetes - Depression	3106	PDC (antidiabetic medication)	PDC < 90%	1614 (52.0%)
Magidson 2012 (USA)[111]	Substance use treatment centre	Cross-sectional (RCT baseline data)	- HIV/AIDS - Substance use	73	ACTG Adherence to Antiretroviral Medication Questionnaire (ART and other medications)	Report mean adherence over previous 4 days	ART: Yesterday = 100%; 2 days ago = 94.6%; 3 days ago = 97.8% 4 days ago = 97.8% Other medications: Yesterday = 97.3% 2 days ago = 94.9% 3 days ago = 96.5% 4 days ago = 97.4%
Manuel 2011 (USA)[112]	Outpatient community mental health centres	Data from RCT	- Psychotic disorder - Active substance use or dependence	198	Self-report missed medication days in past 6 months (psychiatric medications)	≥ 20% missed medication days	30 (15.2%)
Mathew 2013 (India)[113]	Hospital	Non-randomised non-controlled intervention	- Type 2 diabetes - Comorbidity(s) (receiving ≥ 5 drugs)	240	14-item self-report questionnaire (diabetes medication)	Patients were categorised into adherent and non-adherent (criteria for sorting not reported)	116 (48.3%)
May 2010 (USA)[114]	Hospital	Longitudinal observational	- Coronary artery disease - Depression	585; MM = 73	MPR across multiple refills (continuous measure of adherence (CMA) for lipid-lowering medications)	Report mean CMA	Mean adherence: 6 months = 90.9% 1 year = 89.3% 2 years = 85.7%
McCusker 2016 (Canada)[115]	Primary care	Data from RCT	- ≥ 1 chronic physical condition(s) or chronic pain ≥ 6 months - ≥ mild depressive symptoms (PHQ-9)	215; analysed n = 202	MMAS-4 (any medication)	Report mean (SD) MMAS-4 higher score = higher non-adherence (range = 1-4).	Mean (SD) = 0.7 (0.9)

Mellins 2009 (USA)[116]	Community	Cross-sectional (baseline data from cohort study	- HIV - Psychiatric disorder - Substance use disorder	1138 (n prescribed ARV = 542; n prescribed antipsychotics = 501)	Antiretrovirals: Modified version of AACTG Adherence to Antiretroviral Medication Questionnaire Psychiatric medications: 1-item self-report	Self-report < 100% adherence in past 3 days	ARV = 244 (45.0%) Psychiatric medication = 190 (38%)
Miketic 2012 (USA)[117]	Hospital	Prospective descriptive pilot study	- Coronary artery disease - Diabetes mellitus	51; MM = 22	MMAS-4	Report mean (SD)	Mean (SD) = 0.86 (0.94)
Mira 2014 (Spain)[118]	Health centres	RCT	- Multimorbid taking multiple medications	99	MMAS-4	Report mean (SD)	Mean (SD) = 6.9 (1.1)
Monroe 2018 (USA)[119]	HIV clinic	RCT	- HIV - Diabetes and/or hypertension	46	MPR	Report median (IQR)	ART: 92% (71-100%) Other medications: 79% (63-96%) (if patient had hypertension + diabetes, only hypertension adherence recorded)
Moore 2015 (USA)[120]	Ongoing studies at research centre	RCT	- HIV - Bipolar type I or II	50	Visual analogue scale (HIV medication and psychiatric medication)	VAS < 100% = some adherence difficulty in previous 30 days	HIV medication = 12 (47.8%) Psychiatric medication = 14 (54.3%)
Moore 2012 (USA)[121]	Ongoing studies at research centre	Cross-sectional	- HIV - Bipolar type I or II	44 (n with psychiatric medication assessed = 39)	MEMS (psychiatric medication and HIV medication)	MEMS <90%	HIV medication = 23/44 (52.3%) Psychiatric medication = 25/39 (64.1%)
Moore 2012 (USA)[122]	Ongoing studies at research centre	Cross-sectional	- HIV - Current or lifetime history of methamphetamine use/dependence	127; MM = 75	AIDS Clinical Trials Group (ACTG) four day adherence self-report questionnaire (HIV medications)	Report of any skipped dose in previous 4 days	Total = 16 (21.3%) 12 (18%) with lifetime history METH+ use 4 (50%) with current METH+ use
Muntner 2010 (USA)[123]	Community	Cross-sectional	- Chronic kidney disease - Hypertension	13065; MM = 3936	MMAS-4 (antihypertensive medications)	MMAS-4 = 0: 'better adherence' MMAS-4 ≥ 2: worse adherence	MMAS-4 = 0: 2715 (69.0%) MMAS-4 = 1: 916 (23.3%) MMAS-4 ≥ 2: 305 (7.8%)
Muth 2016 (Germany)[124]	General practice	Pilot cluster-RCT	≥ 3 chronic conditions	100	MMAS-4 Medication Adherence Rating Scale (MARS)	MMAS-4: scores 0–4, higher scores = lower adherence	MMAS-4 mean (SD) =0.3 (0.6) MARS mean (SD) = 24.0 (1.4)

					Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	MARS: scores 5–25, lower scores = lower adherence Observed medication adherence: Scores outside 0.8–1.2 are divergent	Observed adherence: Drug score = 43/97 (44.3%) divergent Dose score = 93/94 (98.9%) divergent Regimen score = 93/93 (100%) divergent
Muth 2018 (Germany)[125]	General practice	Pragmatic cluster-RCT	≥ 3 chronic conditions	Total = 505; intervention = 253; control = 252	MMAS-4 Observed medication adherence: (1) drug score (2) dose score (3) the regimen score	Report mean (SD) MMAS-4: scores 0–4, higher scores = lower adherence Observed medication adherence: scores outside 0.8–1.2 are divergent	MMAS-4 =3.7 (0.6) Observed adherence: Drug score = 188/501 (37.5%) divergent Dose score = 259/499 (50.9%) divergent Regimen score = 255/500 (51.0%) divergent
Naidoo 2013 (South Africa)[126]	Primary care	Cross-sectional	- TB - HIV	3107; MM = 1729; analysed n = 757	Anti-TB medication: 1-item self-report of percent medication taken in prior 3–4 weeks HIV medication: 1-item self-report of percent medication taken in prior 4 weeks	Self-report adherence < 90%	TB medication: 315 (41.6%) HIV medication: 321 (42.4%)
Nakimuli-Mpungu 2019 (Uganda)[127]	Primary care	Baseline data from cluster-RCT	- Depression (MINI) - HIV	1140	Missed-dose method: 1-item self-report of days missed taking all doses in prior week (ART)	Self-report adherence< 100%	Not reported
Natarajan 2013 (Canada)[128]	General practice	Cross-sectional	- Diabetes mellitus - Hypertension	527; analysed n = 484	MMAS-4 (antihypertensives)	Answer yes to 1 or more questions = medium or low adherence	Low adherence = 8 (1.7%) Medium adherence = 101 (20.9%) Combined = 109 (22.5%)
Nelson 2011 (USA)[129]	Veterans Affairs medical centre	Retrospective record review	- Hyperlipidaemia - Diabetes - Psychotic disorder (not all sample has psychotic disorder)	total = 124; +psychotic disorder = 62; -psychotic	Cumulative mean gap ratio (CMGR): (both medication classes; for combination therapy, the medication with the greatest non-adherence was used)	Report mean (SD)	Antihyperlipidemic medication: +psychotic disorder mean (SD) = 12.0% (12.2) reflecting approx. 44 days without therapy -psychotic disorder mean (SD) = 16.9% (13.0%) reflecting approx. 62 days without therapy

				disorder = 62			Antipsychotic medication: mean (SD) = 18.3% (16.4%) reflecting approx. 68 days without therapy
Nelson 2011 (USA)[130]	Veterans Affairs medical centre	Retrospective record review	- Diabetes - Schizophrenia or related psychotic disorder	124; MM = 62	Cumulative mean gap ratio (CMGR) (antihyperglycaemic medication; antipsychotic medication)	Report mean (SD)	Antihyperglycaemics = 15.8% (16.8%) Antipsychotics = 18.3% (16.4%)
Neuget 2016 (USA)[131]	Administrative claims database	Retrospective cohort	- Non-metastatic breast cancer - Chronic condition from hypertension, hyperlipidaemia, GERD, thyroid, diabetes, osteoporosis	21255; MM = 13427	MPR (tamoxifen and/or aromatase inhibitor) Non-adherence after breast cancer diagnosis reported per disease group	MPR < 80%	Hypertension group: 1233/8683 (14.2%) Hyperlipidaemia group: 671/5829 (11.51%) GERD group: 567/4264 (13.97%) Thyroid group: 484/3542 (13.67%) Diabetes group: 267/1742 (15.33%) Osteoporosis group: 147/1384 (10.62%)
Newville 2014 (USA)[132]	Methadone maintenance clinic	Cross-sectional	- HIV - Treated with methadone maintenance therapy	100	Self-report using questions from AIDS Clinical Trials Group (HAART)	Self-report < 100%	30 (30%)
Nichol 2009 (USA)[133]	Administrative claims database	Retrospective record review	- Hypertension - Treated with lipid-lowering medication	5943	PDC (antihypertensives; lipid-lowering medication)		Year 1 Non-adherent to at least one = 1817 (30.6%) Year 2 Non-adherent to at least one = 2955 (49.7%)
Otenyo 2018 (Kenya)[134]	Hospital renal unit	Cross-sectional	- Chronic kidney disease - Hypertension	144	MMAS-8 (antihypertensives)	MMAS-8 < 6	61 (42.4%)
Owen-Smith 2016 (USA)[135]	Healthcare database	Retrospective analysis	- Schizophrenia or schizoaffective disorder - Treatment by any statin or ACEI/ARB	1420; MM = 710	MPR (cardiovascular medications; if taking both statin and ACEI/ARB, adherence calculated separately for each)	MPR < 0.80	Statin: 350/583 (60.0%) ACEI/ARB: 232/383 (60.6%)
Padman 2017 (India)[136]	Hospital general medicine department; Dialysis centre	Prospective observational	- Chronic renal failure - Diabetes mellitus	107	MMAS-8 (diabetes medication)	Rated MMAS-8 scores as low, moderate and high adherence (criteria for categorising not reported)	Low = 61 (57.0%) Moderate = 36 (33.6%)

Pantalone 2012 (USA)[137]	HIV/AIDS clinics	Cross-sectional	- HIV - Hepatitis C virus	178; MM = 54; analysed n = 45	1-item self-report number of doses missed in prior 30 days (HIV medications)	Self-report adherence < 100%	17 (37.8%)
Parpouchi 2017 (Canada)[138]	Administrative database	Longitudinal (using baseline data, longitudinal data and retrospective data)	- Mental illness - Treated with methadone maintenance	497; MM with adherence to MMT reported = 78	MPR (methadone)	MRP < 0.8	62 (79.5%)
Patel 2016 (USA)[139]	Medicaid database	Retrospective cohort	- Type 2 diabetes - Developmental disability (cerebral palsy, autism, Down's syndrome, cognitive disabilities)	1529	Adherence: MPRm (anti-diabetes medications) Persistence: period from start of medication therapy to discontinuation (anti-diabetes medications)	Adherence: MPRm < 0.8. Report mean SD. Persistence: gap > 120 days	Pooled mean (SD) MPRm = 0.90 (0.19) Non-persistence: 464 (30.4%)
Pence 2015 (USA)[140]	Infectious disease clinics	Pseudo-cluster RCT	- HIV - Major depression (MINI confirmed diagnosis & PHQ-9)	304	Visual analogue scale (ART adherence in prior 30 days)	Report mean (SD)	Mean (SD) = 86.5% (22.7%)
Pentakota 2013 (USA)[141]	Administrative research database	Retrospective cohort	- Diabetes - Comorbidity(s)	79249; MM at some stage (baseline, end, both) = 71334	Adherence: PDC (diabetes medications) Persistence: time from initiation of treatment to first discontinuation in treatment during first 2 years of treatment (diabetes medications)	PDC < 0.8 Persistence: discontinuity > 60 days	Reported according to concordant and discordant comorbidities: Concordant_Concordant = 34442 (44.5%) Discordant_Discordant = 39469 (50.95%) Additional combinations reported in original paper.
Poon 2009 (USA)[142]	Veterans Health Administration database	Retrospective cohort	- Dementia - Hypertension	56561	MPR (antihypertensive medications)	MPR < 0.8	Reported by race and drug class: Min = 13.0%; Max = 39.6%
Puyat 2017 (Canada)[143]	Healthcare databases	Retrospective cohort	- Depression - Additional chronic health condition	105476; MM = 13329	PDC (antidepressants)	Report mean (SD) in 180-day period and in 365-day period	Mean (SD) = 0.84 (0.24) Mean (SD) = 0.80 (0.27)
Qian 2014 (USA)[144]	Administrative database	Cross-sectional	- COPD - Depression	74863; MM = 25130; analysed n = 12574	PDC (COPD medications) Discontinuation (COPD medications)	PDC: < 0.80 Discontinuation: binary variable of medication stopped	PDC: 8621 (68.6%) Discontinuation: 3009 (23.9%)

						3 months or more prior to end of observation period or death, whichever first	
Rattinger 2012 (USA)[145]	Administrative database	Cross-sectional	- Chronic heart failure - Alzheimer's disease or related disorders	9827; MM = 2379	MPR (chronic heart failure medication)	Report mean (SD)	EBM = 0.86 (0.15) Diuretic = 0.81 (0.20) Cardiac glycoside = 0.86 (0.18) Calcium channel blocker = 0.88 (0.17)
Reed 2018 (Australia)[146]	General practice	RCT	≥ 2 chronic conditions from cardiovascular, respiratory, musculoskeletal, psychological, digestive, kidney disease, diabetes (type 1 or 2), cancer	254	5-item self-report	Higher scores indicate poorer levels of medication adherence (range of possible scores not reported)	Mean (SD) = 24.14 (0.14)
Rosenblum 2014 (USA)[147]	Residential and outpatient psychiatric and substance use facilities	RCT	- Mental illness - History of substance use	203	Medication Adherence Rating Scale (MARS) (psychiatric medication)	Report mean (SD) Lower score = higher medication adherence (range of possible scores not reported)	Mean (SD) = 1.55 (0.48)
Roux 2012 (France)[148]	Hospital	Prospective cohort	- HIV - Hepatitis C virus	593 at enrolment; 246 at follow-up	7-item self-report adherence questionnaire (HIV medications)	Self-report adherence < 100% during 4-days prior	Enrolment: 214 (36%) Follow-up: 72 (29.3%)
Saadat 2015 (Iran)[149]	Healthcare centres	Cross-sectional	- Hypertension - ≥ 1 comorbidity	280; MM = 160	MMAS-8 (Persian version) (antihypertensive medication)	MMAS-8 < 6	77 (48.1%)
Safren 2009 (USA)[150]	Community	RCT	- HIV - Depressive mood disorder	45	MEMS (antiretroviral medication)	Report mean (SD)	Control group = 54.41 (30.94) Intervention group = 62.26 (26.82)
Safren 2012 (USA)[151]	Community outreach, hospital HIV clinics, substance use clinics	RCT	- HIV - History of injection drug use (self-endorsed + enrolled in opioid treatment for ≥ 1 month(s)) - Criteria for diagnosis of current or subsyndromal	89	MEMS (antiretroviral medication)	Report mean (SD) percent MEMS-based adherence since last visit	Mean (SD) = 69.0% (28.3%)

			depressive mood disorder				
Safren 2014 (USA)[152]	Primary care	RCT	- Unipolar depression - Type 2 diabetes	87	MEMS (antidiabetic oral medication, insulin or both)	Report mean (SD)	Control group = 85.7 (24.6%) Intervention group = 76.6% (26.0%)
Santorelli 2016 (USA)[153]	Administrative databases	Retrospective cohort	- Breast cancer - Diabetes or hypertension or lipid disorder	9340; MM groups: diabetes = 298; lipid disorders = 508; hypertension = 1062	Adherence: PDC (hypertension; diabetes; lipid disorder medications) Persistence: largest number of consecutive days without coverage by relevant medication class during first 18 months (hypertension; diabetes; lipid disorder medications)	Adherence: PDC < 80% Persistence: Diabetes = 26 days without medications Hypertension = 26 days without medication Lipid-lowering = 62 days without medications	PDC: Diabetes = 78 (26.2%) Hypertension = 151 (14.2%) Lipid disorder = 147 (28.9%) Persistence: Diabetes = 94 (31.5%) Hypertension = 200 (18.8%) Lipid disorder = 123 (24.2%)
Schmitt 2010 (USA)[154]	Veterans Affairs medical centre	Retrospective observational	- Chronic kidney disease - Treatment with antihypertensive	7227	MPR (antihypertensive medication)	MPR < 80%	2360 (32.7%)
Schüz 2011 (a) (Germany)[155]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring: (1) intentional non-adherence (2) unintentional non-adherence	Report mean (SD) Higher values indicate higher degrees of non-adherence (range of possible scores not reported)	Intentional mean (SD) = 1.32 (0.68) Unintentional mean (SD) = 1.40 (0.67)
Schüz 2013 (Germany)[156]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	215	1-item self-report intentional non-adherence from Reported Adherence to Medications Scale (RAM)	Report mean (SD) Higher scores indicate better adherence (range of possible scores not reported)	Mean (SD) = 4.68 (0.68)
Schüz 2011 (b) (Germany) [157]	Community	Longitudinal	≥ 2 chronic conditions from Charlson Comorbidity Index or	309	2 self-report items from Reported Adherence to Medication Scale (RAM) measuring:	Not reported	Not reported

			Functional Comorbidity Index		(1) intentional non-adherence (2) unintentional non-adherence		
Shi 2010 (USA)[158]	Veterans Health administration database	Retrospective analysis	- Schizophrenia - Metabolic syndrome	593; MM = 209	Adherence: MPR (antipsychotics) Discontinuation: medication gap (antipsychotics)	Adherence: report mean (SD) MPR Discontinuation: medication gap > 30 consecutive days	Mean (SD) MPR = 0.33 (0.47) Median time to discontinuation = 50 days
Shin 2010 (South Korea) [159]	Pharmacy claims data	Retrospective record review	- Hypertension - Cancer survivor	2455193; MM = 12636	CMA (antihypertensive medication)	CMA < 80%	5762 (45.6%)
Shoval 2017 (Israel)[160]	Healthcare database	Retrospective analysis	- Parkinson's disease - Depression	8553	MPR (antidepressant medication)	MPR < 20% = non-adherent MPR 20% - 50% = poorly adherent MPR 50% - 80% = moderately adherent	< 20% = 1566 (18.4%) 20-50% = 1184 (13.8%) 50-80% = 1584 (18.5%) < 80% = 4334 (50.7%)
Siika 2013 (Kenya)[161]	Ambulatory HIV care program	Retrospective cohort	- HIV - Tuberculosis	21242; MM = 5186; analysis n = 4838	1-item self-report number of pills taken during prior week (ART)	If patient ever answers 'most', 'half', 'few', or 'none' (defined as non-perfect adherence to ART)	986 (20.4%)
Slabbert 2015 (South Africa)[162]	Administrative database	Prospective cohort	- Major depressive disorder - HIV/AIDS	127	MPR (antidepressant medications)	Report mean (SD)	74.43 (32.03)
Soto 2015 (Chile) [163]	Hospital	Prospective	- Type 2 diabetes - Hypertension	50	Pill count (seems like both medications together)	Pill count < 80% or > 110%	35 (70.0%)
Stack 2010 (UK)[164]	Healthcare database (secondary care)	Cross-sectional	- Type 2 diabetes - Hypertension	480	MMAS-4 (separately for antidiabetic medication, antihypertensive medication and statin)	0 = total adherence 4 = non-adherence: presented as no unintentional non-adherence; no intentional non-adherence	Unintentional non-adherence: Diabetes medication = 302 (62.9%) Hypertension medication = 226 (47.1%) Statins = 225 (46.9%) Intentional non-adherence: Diabetes medication = 63 (13.1%)

							Hypertension medication = 92 (19.2%) Statins = 87 (18.1%)
Stilley 2010 (USA)[165]	Primary care	RCT	≥ 2 from type 2 diabetes, hypertension, hyperlipidaemia	373	MEMS (not clear to what medications)	Report mean (SD): (1) prescribed number of doses (2) days with prescribed dose (3) days with correct inter-dose interval	Prescribed number of doses = 92.6 (17.6) Days with prescribed dose = 82.2 (22.4) Days with correct inter-dose interval = 64.2 (30.1)
Stuart 2015 (USA)[166]	Medicare database	Retrospective cohort	- Diabetes - Cancer	32855; MM = 4348	PDC (diabetes medication)	Report mean PDC	OHA (n = 2248) mean PDC = 73.8% RAAS-I (n = 2618) mean PDC = 70.5% Statins (n = 2272) mean PDC = 69.0%
Taira 2017 (USA)[167]	Healthcare database	Retrospective observational	- Type 2 diabetes - Treatment with antihypertensive, oral antidiabetic and cholesterol lowering medication	5163	MPR (each medication)	MPR < 0.8	Range of non-adherence reported: 16% - 48.2% Non-adherence for each medication according to ethnicity reported in original paper.
Tang 2014 (USA)[168]	Outpatient sites	Secondary analysis of prospective descriptive cohort	- Heart failure - Depression (GDS)	280; MM = 103	MEMS (heart failure medication); Basel Assessment of Adherence Scale (BAAS) (heart failure medication)	MEMS: < 80% BAAS: a positive answer on any question	Self-report = 77 (75%) MEMS = 29 (28%)
Thaler 2015 (USA)[169]	HIV treatment facilities	Not reported	- HIV - neurocognitive impairment and/or substance dependence/use and/or psychiatric illness	204; MM not reported	MEMS (HAART) Self-report (not described) (HAART)	Report mean (SD) per additional risk factor	One risk factor: 64.4% (27.0%) Two risk factors: 62.8% (28.0%)
Vacek 2013 (USA)[170]	Medicaid database	Retrospective analysis	- Developmental disability - Hypertension	280; analysed n = 201	PDC (antihypertensive medication)	PDC < 0.8	90 (45%)
Vatcharavongvan 2017 (Thailand)[171]	Primary care	Cross-sectional	≥ 2 chronic conditions from type 2 diabetes, hyperlipidaemia,	397; analysed n (polyphar	Structured questionnaire during face-to-face interview	At least one medication that was (a) taken at wrong	74 (60.7%)

			hypertension, ischemic heart disease, chronic heart failure, cerebrovascular disease, epilepsy	macy group) = 122		time, (b) missed (c) intentionally stopped by patient during 2 weeks prior	
Vega 2017 (USA)[172]	Healthcare database	Retrospective cohort	- Major depressive disorder - Type 2 diabetes	1361	Adherence: MPR (depression medication; diabetes medication) Persistence: Length of treatment (LOT) (depression medication)	MPR < 80% LOT ≥ 15 day gap in first 180 days of therapy	Depression medication: Non-adherent = 872 (64.1%) Non-persistent = 926 (68.0%) Non-adherent/non-persistent = 936 (68.8%) Diabetes medication: Non-adherent = 525 (38.6%)
Villar 2009 (Spain)[173]	Ongoing studies, administrative database	Prospective cohort	- Severe obstructive sleep apnoea - Cardiovascular disease	2158; antihypertensives = 659; statins = 175; antiplatelets = 109	MPR (cardiovascular disease medications)	MPR < 80%	Antihypertensives: 118/659 (17.91%) Statins: 59/175 (33.71%) Antiplatelets: 9/109 (8.26%)
Wagner 2011 (USA)[174]	Pooled data from longitudinal studies	Cross-sectional and longitudinal	- HIV - Depression (BDI-II; BDI (original version); CESD; depression subscale of Brief Symptom Inventory)	1374; MM = 489	MEMS (HIV medication)	MEMS < 90%	Severe depression = 72% (148/205) Mild to moderate depression = 56% (159/284) Combined = 307/489 (62.8%)
Wakefield 2012 (USA)[175]	Primary care	RCT	- Type 2 diabetes - Hypertension	302 (control = 107; low intensity = 102; high-intensity = 93)	1-item self-report adherence	Report mean (SD)	Mean (SD): Control group = 99.5 (4.4) Low-intensity group = 99.7 (1.6) High-intensity group = 99.7 (2.7)
Warner 2013 (Germany)[176]	Community	Longitudinal	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	309	1-item self-report unintentional non-adherence from Reported Adherence to Medication Scale (RAM)	1 < RAM < 4	T1 = 83 (27%) T2 = 99 (32%)
Wei 2018 (USA)[177]	Medicare database	Retrospective cohort	- Major depression - Chronic obstructive pulmonary disease	25458; analysed n = 11572	PDC (COPD medications)	PDC < 0.8	7059 (61.0%)

Weiss 2016 (USA)[178]	Ambulatory practices	Longitudinal	- HIV - Hypertension or chronic kidney disease	152; +hypertension = 115 (analysed n = 102); +CKD = 37 (analysed n = 20)	MEMS (1) taking adherence (2) dosing adherence (3) timing adherence (HIV medications and CKD/hypertension medications separately)	MEMS < 95% (HIV medications) MEMS < 80% (other medications)	+hypertension: HIV medication = 40 (39%) Hypertension medication = 20 (20%) +CKD: HIV medication = 7 (35%) CKD medication = 4 (20%)
Wenze 2015 (USA)[179]	Hospital	Pilot RCT	- Bipolar I or II - Substance use disorder	30	Adapted version of Medication Compliance Questionnaire (MCQ) (prescribed mood-stabilizing agent)	MCQ ≥ 2	12 (40.0%)
Wong 2014 (China)[180]	Administrative database	Cross-sectional	≥ 2 chronic conditions	3866; MM = 523	MMAS-8 (antihypertensives, oral hypoglycaemic agents, lipid lowering agents separately)	MMAS-8 ≤ 6 (MMAS-8 sections for each drug should score >6 to be defined as having optimal adherence)	321 (45.1%)
Wroe 2014 (Rwanda)[181]	Rural health centres	Cross-sectional	- HIV - Depression (Hopkin's Symptom Checklist)	292; MM = 32	Self-report 4-day recall (HIV medications) Self-report 2-day and 7-day treatment interruptions (HIV medications) CASE index (HIV medications)	Self-report 4-day recall: ≥ 1 dose(s) in four days prior Self-report 2-day and 7-day treatment interruptions: non-adherence on sequential days CASE index ≤ 9	4-day recall = 5 (17.0%) 2-day interruptions = 7 (22.2%) CASE index = 5 (15.8%)
Wu 2011 (USA)[182]	Medicaid database	Retrospective observational	- Hyperlipidaemia - Type 2 diabetes	1705	MPR (statins; antidiabetic medication)	MPR < 0.8	Statins = 1079 (63.3%) Mean (SD) MPR (statins) = 0.61 (0.32) Mean (SD) MPR (antidiabetics) = 0.71 (0.30)
Wu 2010 (USA)[183]	Medicaid database	Retrospective cohort	- Major depressive disorder - Anxiety disorder	3083; MM = 776	Adherence: MPR modified (MPRm) (antidepressants) Persistence: Refill gap (antidepressants)	MPRm: report mean (SD) Refill gap: median (IQR)	Mean (SD) MPRm = 0.71 (0.37) Median (IQR) persistence (days) = 237 (216-256)

Wu 2013 (USA)[184]	Inpatient and outpatient cardiology clinics	Secondary analysis of two prospective studies	- Heart failure - Depression (PHQ-9)	216; MM = 55	MEMS (heart failure medication)	MEMS < 88%	30 (54.55%)
Wulaningsih 2018 (Sweden)[185]	Administrative database	Retrospective analysis	- ER-positive breast cancer - Charlson Comorbidity Index score ≥ 1	4645; MM baseline = 715; MM follow-up = 1348	MPR (endocrine therapy)	MPR < 80%	Baseline: 168/715 (23.5%) 5-years: 330/1348 (24.5%)
Xia 2015 (China)[186]	Hospital	Cross-sectional	- ≥ 1 chronic condition from chronic obstructive pulmonary disease, diabetes, hyperlipidaemia, hypertension - Depression (PHQ-9) and/or anxiety (GAD-7)	325; +anxiety = 276; +depression = 266	Chinese Medication Adherence Scale (CMAS) (chronic condition)	CMAS <6 to	+depression: 222 (83.5%) +anxiety: 226 (81.9%)
Xue 2018 (China)[187]	Primary care	Baseline data from RCT	- Hypertension - Depression (PHQ-9)	2362	1-item self-report adherence (antihypertensive medications)	Answer yes to "do you sometimes forget to take your antihypertensive pills"	954 (40.4%)
Yang 2010 (USA)[188]	Medicare database	Retrospective cohort	- Diabetes - Hypertension and/or renal disease	599141	PDC (antihypertensive medication)	PDC < 0.8	276945 (46.22%)

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